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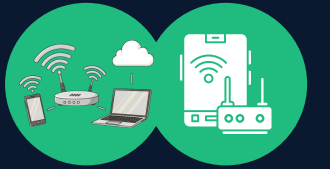
Hacking Wireless Networks

ETHICAL HACKING AND PENETRATION TESTING
CCS6324 - 2530

Dr. Hafiz Adnan Hussain

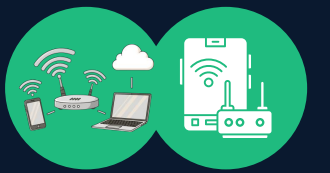


DISCLAIMER



- This course is designed to educate students on ethical hacking principles. The knowledge and techniques acquired must be used responsibly and not for any illegal or unauthorized activities.
- Each student is solely responsible for their actions.
- Do not investigate individuals, websites, servers or conduct any illegal activities on any system you do not have permission to analyze.
- The course content has been developed in good faith, drawing from the author's experience and reputable sources. While every effort has been made to ensure accuracy, the author does not guarantee absolute correctness. Students are encouraged to exercise discretion and actively engage in discussions by raising any questions or concerns during the course.

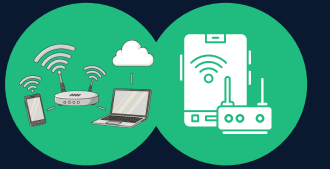
Learning Objectives



What You Will Learn?

- Wireless technology fundamentals and IEEE 802.11 standards
- Evolution of wireless security protocols (Wired Equivalent Privacy - WEP → WPA3)
- Identify authentication and association processes in wireless networks
- Demonstrate wireless reconnaissance and wardriving techniques
- Execute wireless penetration testing attacks ethically
- Design and implement wireless security countermeasures
- Analyze real-world wireless breach case studies

Module 1: Wireless Technology Fundamentals

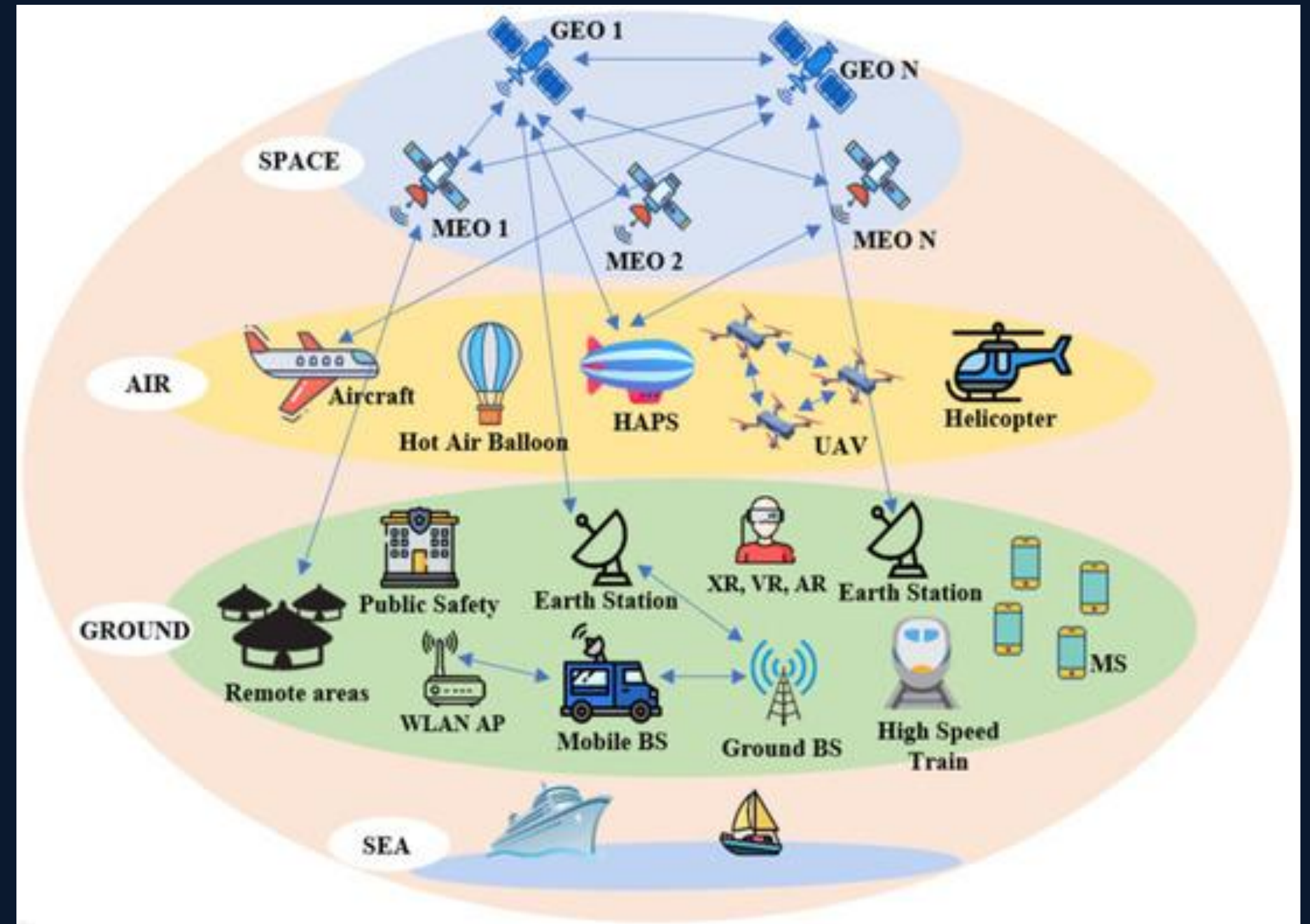


1.1 What is Wireless Technology?

Wireless technology enables communication without physical cable connections. In modern networks, wireless is ubiquitous.

Wireless technology is part of our lives:

- IoT devices (smart homes, sensors, baby monitors)
- Enterprise access points
- Keyless entry systems
- Cellphones and mobile devices (smartphones, tablets)
- GPS
- Satellite communications
- Cellular networks (4G/5G/6G)
- Remote controls
- Garage door openers
- Two-way radios
- Bluetooth speakers



This module focuses on Wireless Local Area Networks (WLANs) using IEEE 802.11 standards, commonly known as Wi-Fi.

Module 1: Wireless Technology Fundamentals (Cont'd)

1.2 Wireless Network Components

Every wireless network requires three fundamental components:

Access Point (AP)

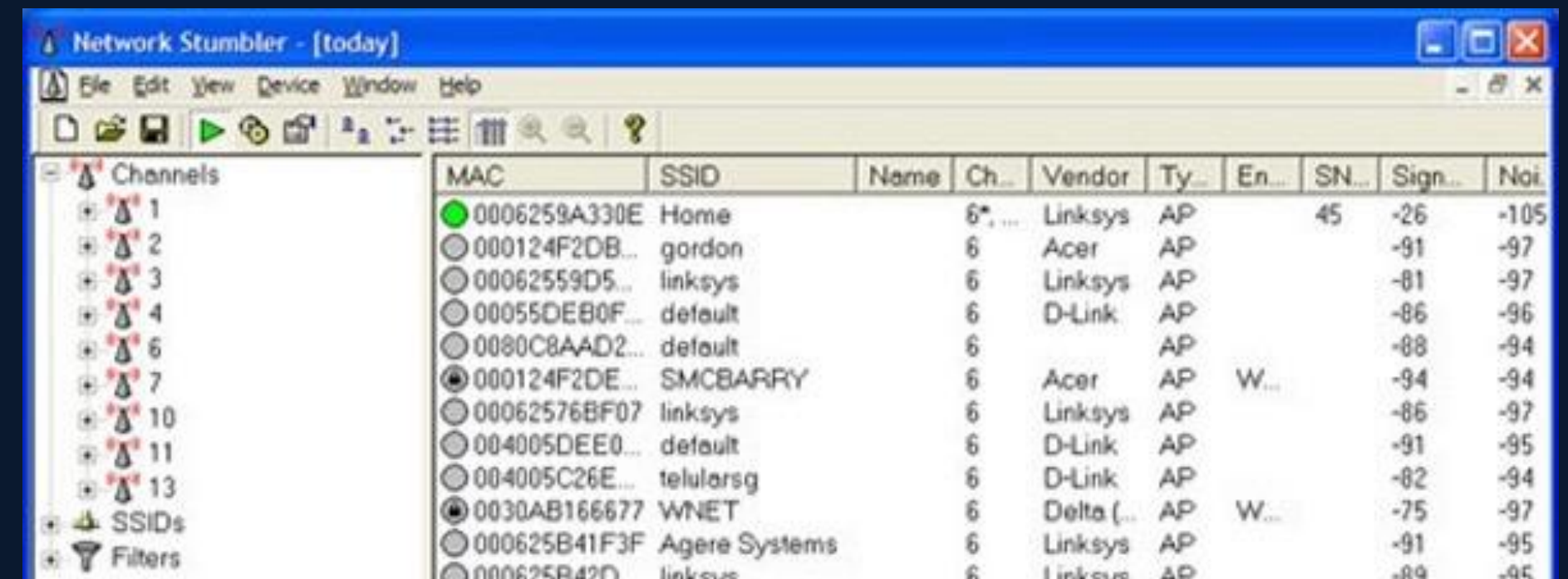
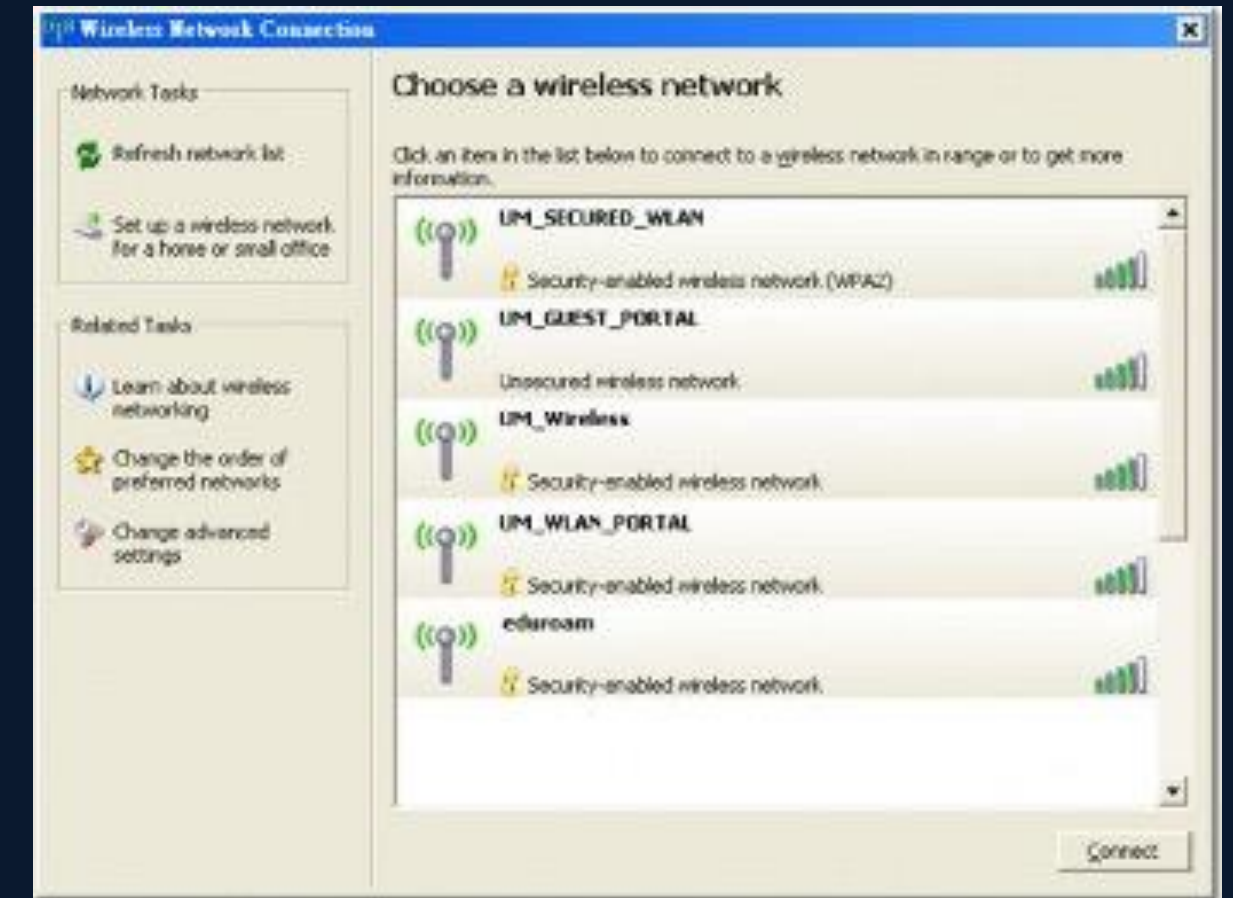
- Acts as a transceiver (transmitter + receiver)
- Connects to a wired Ethernet backbone
- Creates a coverage area (cell)
- Manages client associations
- Bridges wireless clients to the wired LAN

Wireless Network Interface Card (WNIC)

- Hardware adapter in client devices
- Converts radio signals to digital data
- Operates in different modes (managed, monitor, promiscuous)
- Contains the MAC address for identification

Service Set Identifier (SSID)

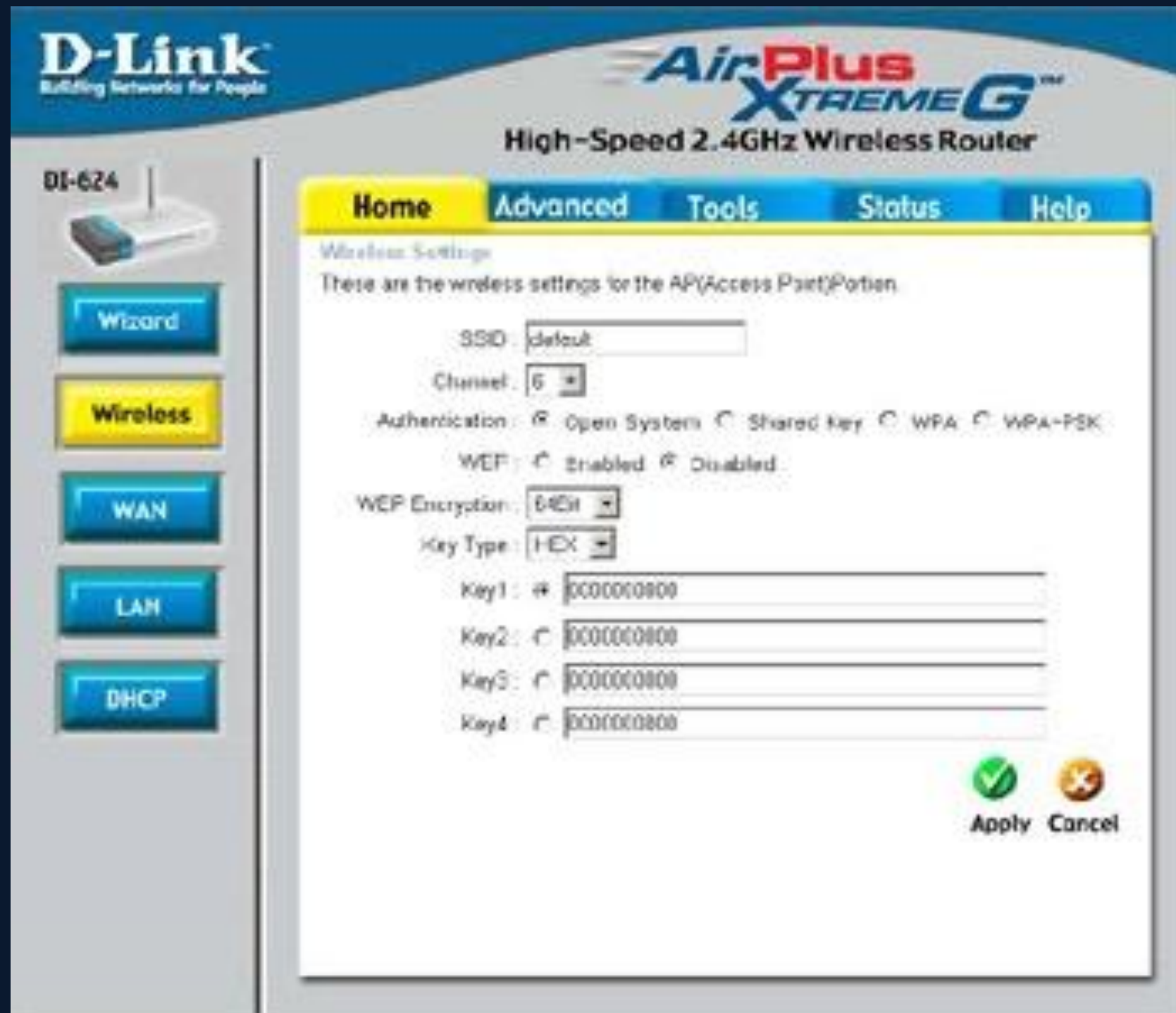
- Network name (1-32 alphanumeric characters, case-sensitive)
- Broadcast by AP (can be hidden)
- Used by clients to identify and connect to networks
- Not a security measure despite the common misconception



Module 1: Wireless Technology Fundamentals (Cont'd)

Wireless Router & Configuration Options

A wireless router includes an access point, a router, and a switch. Configuration options: change SSID, Wired Equivalent Privacy (WEP) encryption, WiFi Protected Access (WPA) is better, and Turn off SSID broadcast.



D-Link Building Networks for People
AirPlus Xtreme G™
High-Speed 2.4GHz Wireless Router

DI-624

Home **Advanced** Tools Status Help

Wireless Settings
These are the wireless settings for the AP(Access Point)Portion.

SSID: default

Channel: 6

Authentication: ☒ Open System ☐ Shared Key ☐ WPA ☐ WPA-PSK

WEP: ☐ Enabled ☒ Disabled

WEP Encryption: 64Bit

Key Type: HEX

Key1: # 0000000800

Key2: 0000000800

Key3: 0000000800

Key4: 0000000800

Apply Cancel

Wizard
Wireless
WAN
LAN
DHCP



D-Link Building Networks for People
AirPlus Xtreme G™
High-Speed 2.4GHz Wireless Router

DI-624

Home **Advanced** Tools Status Help

Wireless Performance
These are the Wireless Performance features for the AP(Access Point) Portion.

TX Rates: Auto

Transmit Power: 100%

Beacon interval: 100 (msec, range:20~1000, default:100)

RTS Threshold: 2346 (range: 256~2346, default:2346)

Fragmentation: 2346 (range: 1500~2346, default:2346, even number only)

DTIM interval: 1 (range: 1~255, default:1)

Preamble Type: ☒ Short Preamble ☐ Long Preamble

SSID Broadcast: ☒ Enabled ☐ Disabled

802.11g Only Mode: ☐ Enabled ☒ Disabled

CTS Mode: ☐ None ☐ Always ☒ Auto

Super G Mode: Disabled

Disabled
Super G without Turbo
Super G with Dynamic Turbo
Super G with Static Turbo

Apply Cancel Help

Virtual Server
Applications
Filters
Parental Control
Firewall
DMZ
Performance

Module 1: Wireless Technology Fundamentals (Cont'd)

1.3 Wireless Network Modes

Open Mode

- No authentication required
- AP accepts all association requests
- Vulnerable to eavesdropping and unauthorized access
- Still used in public spaces (cafes, airports)

Secure Mode

- Requires authentication (credentials, certificates, or pre-shared key)
- Implements encryption protocols (WEP, WPA, WPA2, WPA3)
- Validates device legitimacy before association
- Recommended for all private networks

Module 1: Wireless Technology Fundamentals (Cont'd)

1.4 Understanding Wireless Network Standards

A standard is a set of rules formulated by an organization: Institute of Electrical and Electronics Engineers (IEEE), which defines several standards for wireless networks.

IEEE Standards

Standards pass through these groups:

- Working group (WG)
- Sponsor Executive Committee (SEC)
- Standards Review Committee (RevCom)
- IEEE Standards Board

IEEE Project 802 to create

- LAN and WAN standards

Module 1: Wireless Technology Fundamentals (Cont'd)

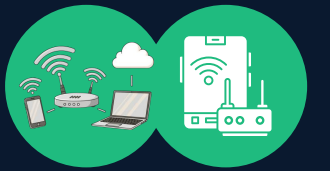
1.4 IEEE 802.11 Standards Overview

Standard	Release	Frequency	Speed	Key Features
802.11a	1999	5 GHz	54 Mbps	Less interference, shorter range
802.11b	1999	2.4 GHz	11 Mbps	Longer range, affordable
802.11g	2003	2.4 GHz	54 Mbps	Backward compatible with 11b
802.11n	2009	2.4/5 GHz	600 Mbps	MIMO, multiple antennas
802.11ac	2013	5 GHz	1.3 Gbps	Wide bandwidth (80/160 MHz)
802.11ax (WiFi 6)	2019	2.4/5 GHz	9.6 Gbps	OFDMA, improved efficiency
802.11be (WiFi 7)	2024	2.4/5/6 GHz	46 Gbps	Multi-link operation, low latency

Current Frequency Bands:

- 2.4 GHz: 11 channels (overlap issues), worldwide availability
- 5 GHz: 23+ non-overlapping channels, less interference, shorter range
- 6 GHz (Wi-Fi 6E): 14+ channels, higher throughput, new regulations

Module 2: Wireless Authentication & Association



2.1 The Association Process

Before data transmission, clients must complete the association:

1. Scanning Phase: Client searches for available networks (passive or active scan)
2. Authentication Frame: Client requests authentication using 802.11 authentication
3. Association Request: Client provides capabilities, supported rates
4. Association Response: AP confirms association, assigns association ID
5. Security Handshake: 4-way handshake (WPA/WPA2) or direct connection (WEP/open)

2.2 802.1X Port-Based Network Access Control

Concept: All ports are blocked until authentication succeeds.

Components:

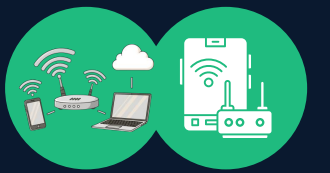
- Supplicant: Client requesting network access
- Authenticator: AP enforcing access control
- Authentication Server: RADIUS server validating credentials

Process:

1. Client connects to AP (unauthenticated)
2. AP blocks non-EAP traffic
3. Client sends EAP identity
4. AP relays to the RADIUS server
5. Server validates credentials
6. Upon success, AP opens the port for the authenticated client

Advantages: Centralized authentication, certificate support, mutual authentication

Module 3: Wireless Security Protocols Evolution



3.1 Wired Equivalent Privacy (WEP) - DEPRECATED

Introduced: 1999 (IEEE 802.11b)

Status: Completely broken, must not be used

Design Intent: Match the security of wired networks

Technical Details:

- Encryption: RC4 stream cipher
- Key Size: 40-bit or 104-bit (plus 24-bit IV)
- IV (Initialization Vector): 24-bit, sent in plaintext
- Authentication: Shared key authentication (optional)

Critical Vulnerabilities:

IV Reuse Weakness

- After $\sim 2^{12}$ packets (4,096), IV typically repeats
- When IV repeats with the same key, an identical keystream is generated
- An attacker can XOR two ciphertexts to eliminate the key and recover the plaintext

Critical Vulnerabilities (Cont'd):

RC4 Weakness

- Stream cipher reuse compromises security
- Biased keystream generation in RC4
- Can recover the key from sufficient keystream samples

ARP Replay Attack

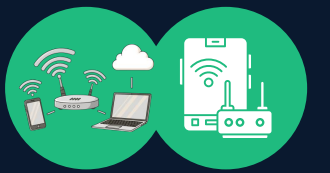
- Attacker captures ARP packets (broadcast, predictable)
- Injects forged ARP requests repeatedly
- Forces AP to respond with encrypted ARP replies
- Quickly exhausts IV space (64K packets = ~ 1 minute)
- The attacker collects enough different IVs to crack the WEP key

Lack of Integrity Protection

- An attacker can flip bits in encrypted packets
- CRC-32 checksum is linear (attacker can adjust the checksum)
- Enables packet injection attacks

Cracking Time: Modern tools crack WEP in minutes with moderate traffic

Module 3: Wireless Security Protocols Evolution (Cont'd)



3.2 Wi-Fi Protected Access (WPA) - LEGACY

Introduced: 2003 (interim solution to WEP issues)

Status: Deprecated but sometimes required for legacy devices

Improvements Over WEP:

- Temporal Key Integrity Protocol (TKIP) for per-packet key generation
- Message Integrity Check (MIC) for replay detection
- Extended Initialization Vector (IV) with sequencing
- Dynamic key rekeying

TKIP Enhancements:

1. Per-Packet Key Mixing: Uses MAC addresses to vary the key per packet
2. MIC (Message Integrity Check): Prevents packet injection
3. IV Sequencing: Detects and discards out-of-order packets
4. Key Rotation: Keys change regularly, not reused

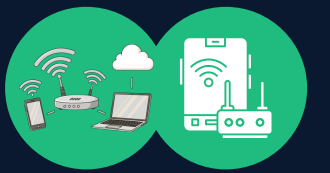
WPA Modes:

- WPA-PSK: Pre-shared key (personal networks)
- WPA-Enterprise: 802.1X with RADIUS (corporate networks)

Remaining Vulnerabilities:

- Fragmentation attacks enable packet injection despite MIC
- TKIP is fundamentally weak as a patch over RC4
- Still vulnerable to dictionary attacks on PSK
- De-authentication frame is unencrypted, enabling DoS

Module 3: Wireless Security Protocols Evolution (Cont'd)



3.3 WPA2 (802.11i Standard) - CURRENT MAINSTREAM

Introduced: 2004 (full standard implementation)

Status: Widely deployed, but being phased out for WPA3

Major Improvements:

- Advanced Encryption Standard (AES) replacing RC4
- Counter Mode with CBC-MAC (CCMP) for confidentiality and authenticity
- Stronger key derivation using PBKDF2
- Mandatory for Wi-Fi certification since 2006

WPA2-PSK 4-Way Handshake:

Step 1: AP → Client sends ANonce (random)

Step 2: Client generates PTK from PSK, derives MIC, sends SNonce

Step 3: AP generates PTK, sends GTK + MIC

Step 4: Client acknowledges with an empty frame

Where:

- $PMK = PBKDF2(\text{passphrase}, \text{SSID}, 4096 \text{ iterations})$
- $PTK = PRF(PMK, \text{"Pairwise key expansion"}, \min(\text{AP_MAC}, \text{STA_MAC}) + \max(\text{AP_MAC}, \text{STA_MAC}) + \min(\text{ANonce}, \text{SNonce}) + \max(\text{ANonce}, \text{SNonce}))$
- $MIC = HMAC\text{-}MD5(PTK[\text{key_mic}], \text{message})$

Known Attacks:

1. KRACK Attack (Key Reinstallation Attack) - CVE-2017-13077

- Exploits WPA2 4-way handshake
- Attacker forces a replay of Message 3
- Client reinstalls PTK, resetting nonce/replay counter to 0
- Encrypted traffic sent with reused nonce
- AES-CCMP with nonce reuse = keystream recovery
- Impact: Decrypt and forge packets on affected networks

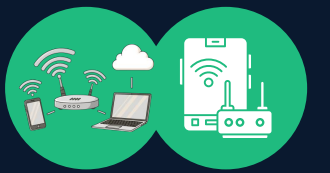
2. Dictionary/Brute Force Attack

- Attacker captures 4-way handshake
- Offline: tries password dictionary against PBKDF2
- If a match is found, it derives the PMK and verifies it against the captured MIC
- Weak passwords are cracked in hours
- No online rate limiting

3. PMKID Hashcat Attack (CVE-2022-6031)

- Captures PMK Identifier from Association Request
- Performs offline PBKDF2 crack against PMKID
- No full handshake needed, speeds up attacks
- Requires monitor mode capture

Module 3: Wireless Security Protocols Evolution (Cont'd)



3.4 WPA3 - LATEST STANDARD

Introduced: 2018, mandatory from 2020 for Wi-Fi 6+ devices

Status: Modern security, gradual adoption

Security Enhancements:

1. Simultaneous Authentication of Equals (SAE)

- Replaces Pre-Shared Key (PSK) authentication
- Differential privacy property: even weak passwords resisted online dictionary attacks
- Mutual authentication: prevents evil twin spoofing
- Eliminates offline dictionary attacks (no way to verify a guess without an interactive protocol)

1. Protected Management Frames (PMF)

- Encrypts deauthentication and disassociation frames
- Prevents deauth-based denial of service
- Prevents SSID broadcast interception

1. Opportunistic Wireless Encryption (OWE)

- Encryption on open networks (no authentication)
- Prevents passive eavesdropping on public Wi-Fi
- Device-to-AP tunnel only

1. 192-bit Encryption (WPA3-Enterprise)

- Uses CNSA Suite (Commercial National Security Algorithm)
- 192-bit encryption for government/sensitive networks

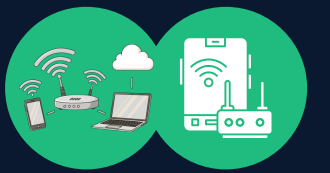
Module 3: Wireless Security Protocols Evolution (Cont'd)



3.5 Security Comparison Table

Feature	WEP	WPA	WPA2	WPA3
Encryption	RC4	TKIP	AES	AES-256
Key Length	104-bit	128-bit	128-bit	192-bit
IV Size	24-bit	48-bit	128-bit	128-bit
Authentication	Shared Key	PSK/802.1X	PSK/802.1X	SAE/802.1X
Replay Protection	No	Sequencing	Counter	Counter
Integrity	CRC-32	MIC	MIC	MIC
Status	Broken	Deprecated	Current	Emerging

Module 4: Wireless Reconnaissance & Wardriving



4.1 Wardriving Concept

Definition: Driving through areas with wireless networking equipment to discover, map, and analyze available wireless networks.

Legal Status:

- Passive discovery is generally legal
- Accessing networks without authorization is illegal
- Wardriving reveals vulnerabilities for penetration testing (with authorization)

4.2 Passive vs. Active Scanning

Passive Scanning:

- Device listens to beacon frames (AP broadcasts every 100ms)
- Captures SSID, MAC, channel, signal strength
- No probe frames sent, difficult to detect
- Works in monitor mode on WNIC

Active Scanning:

- Client sends probe requests on each channel
- AP responds with a probe response
- Faster discovery, but generates traffic
- May trigger intrusion detection systems

4.3 Information Gathered During Reconnaissance

1. SSID: Network name (broadcast or hidden)
2. MAC Address: Unique AP identifier (OUI reveals manufacturer)
3. Channel: 2.4 GHz (1-13) or 5 GHz (36+)
4. Signal Strength (RSSI): -30 dBm (strong) to -90 dBm (weak)
5. Encryption Type: WEP, WPA, WPA2, WPA3, or Open
6. Encryption Level: Not always visible
7. Hidden SSID: Indicated by lack of name broadcast
8. Band Width: 20/40/80/160 MHz

Module 4: Wireless Reconnaissance & Wardriving (Cont'd)

4.4 Common Wardriving Tools

Kismet-ng (Linux, Mac, Windows)

- Advanced packet sniffer and network detector
- Real-time GPS mapping
- Detects hidden SSIDs (through client probe requests)
- Identifies rogue APs through behavior analysis
- Logs pcap files for offline analysis

Aircrack-ng Suite (Linux)

- airodump-ng: Network discovery and packet capture
- airmon-ng: WNIC mode management
- aircrack-ng: WEP/WPA cracking engine
- Aircrack-ng: Packet injection and replay attacks

Wireshark (Cross-platform)

- Deep packet analysis
- Protocol dissection
- Real-time or pcap file analysis
- Filter capabilities for traffic examination

WiFi Analyzer / WiFi Inspector (Mobile/Desktop)

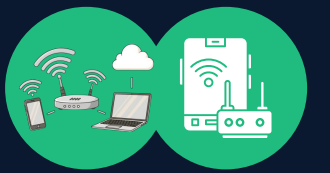
- Consumer tools for signal visualization
- Channel overlap detection
- Network health monitoring

4.5 Identifying Vulnerable Networks

Red flags indicating weak security:

- Open WLAN (no encryption)
- WEP Encryption (completely broken)
- Hidden SSID (false sense of security)
- Default SSID (indicates default credentials likely unchanged)
- Low Signal Strength (indicates AP location)
- WPA2-PSK with weak password (attackable)
- MAC Filtering Only (trivial to spoof)

Module 5: Wireless Attack Techniques



5.1 Evil Twin / Rogue Access Point Attack

Concept: Attacker creates a fake AP with a legitimate SSID to intercept traffic

Execution Steps:

- Identify target SSID and security settings (open networks are easier)
- Configure the attacker's laptop as an AP using hostapd or similar
- Broadcast the same SSID, with an optionally stronger signal than the legitimate AP
- Disable WPA/encryption or use a known password
- Clients connect to rogue AP
- Attacker performs a man-in-the-middle attack or harvests credentials

Impact:

- Credential harvesting (SSL stripping removes HTTPS)
- Session hijacking
- Malware distribution
- Traffic inspection

Real Example: Starbucks "Free Wifi" spoofing has affected thousands of users connecting to fake networks

5.2 De-authentication Attack

Concept: Force client disconnection, capture re-association/handshake, or cause denial of service

Mechanism:

- De-authentication frames (type 0) are unencrypted even in WPA2
- Attacker forges frame: source = AP MAC, destination = client MAC
- Client receives a disconnect, must reassociate
- WPA3 prevents this by encrypting management frames

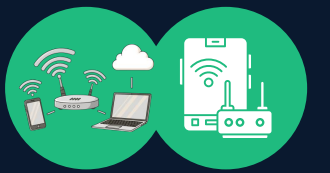
Attack Chain:

1. Monitor for clients connected to AP
2. Send de-auth to a specific client
3. Client disconnects and roams
4. Client attempts reconnection
5. Capture 4-way handshake
6. Offline password cracking

Countermeasures:

- 802.11w (Management Frame Protection) in WPA2
- Mandatory in WPA3
- Not enabled by default on many legacy devices

Module 5: Wireless Attack Techniques (Cont'd)



5.3 Dictionary/Brute Force Attack on WPA2-PSK

Prerequisites:

- 4-way handshake captured (pcap file)
- Target SSID known

Attack Process:

1. Extract ANonce, SNonce, and MIC from handshake capture.
2. For each password in the dictionary:
 - Derive PMK = PBKDF2(password, SSID, 4096 iterations)
 - Derive PTK using nonces and MAC addresses
 - Calculate MIC
 - Compare with captured MIC
3. On match: password found
4. Decrypt captured traffic using the derived key

Tools:

- aircrack-ng: CPU cracking
- hashcat: GPU acceleration (100x faster)
- Rainbow tables: Pre-computed hashes (requires storage)

Example Timeline:

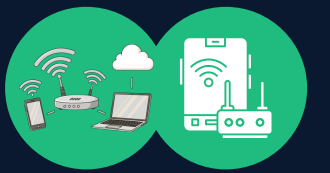
- 8-character password: ~30 minutes (GPU)
- 12-character password: ~8 hours (GPU)
- 20-character random: $\sim 10^{26}$ possibilities (computationally impossible)

Mitigation:

- Use a strong passphrase (20+ characters)
- Use WPA2/WPA3-Enterprise with 802.1X

Avoid dictionary words, use mixed case + numbers + symbols

Module 5: Wireless Attack Techniques (Cont'd)



5.4 PMKID Attack (CVE-2022-6031)

Advantage Over Handshake Capture:

- No full 4-way handshake needed
- Extracted from a single Association Request
- Speeds up the attack significantly

Extraction Process:

1. Listen for Association Request frames in monitor mode
2. Extract PMKID from RSN Information Element
3. Save PMKID and SSID

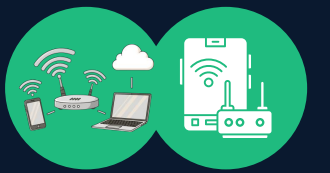
Offline Cracking:

- Attempt PBKDF2 against PMKID
- Faster verification than a full handshake MIC check
- Same password results

Tools:

- hcxdumpool: PMKID extraction
- hashcat: PMKID cracking

Module 6: Case Studies & Real-World Breaches



6.1 WEP Cracking (Historical - 2004)

Context: WEP was the default on all consumer routers until 2006

Attack Execution:

1. Passive listening gathers ~500K packets (ARP requests)
2. Active injection of forged ARP requests accelerates IV exhaustion
3. Collection of unique IVs with RC4 keystream samples
4. Statistical analysis recovers the WEP key in <5 minutes

Impact: Thousands of networks compromised, demonstrating the feasibility of wireless attacks

6.2 KRACK Attack (2017)

Discovery: Mathy Vanhoef and Frank Piessens, University of Leuven

Affected: All WPA2 implementations (vendor-specific variations)

Attack Details:

- Exploits 4-way handshake retransmission logic
- Forces the client to reinstall PTK with the replayed nonce
- Enables decryption of data frames post-handshake
- Additional impact on some implementations: nonce reset to 0

Vendors Affected:

- Android 6.0: Complete key reinstallation, all traffic decryptable
- macOS/iOS: Partial impact (key not reset)
- Windows: Limited impact (only ARP traffic)
- Linux (wpa_supplicant): Complete impact on older versions

Resolution: 802.11w Management Frame Protection makes an attack infeasible

Module 6: Case Studies & Real-World Breaches (Cont'd)

6.3 T-Mobile 2022 Wireless API Breach

Incident: 54 million customer records exposed

Attack Vector:

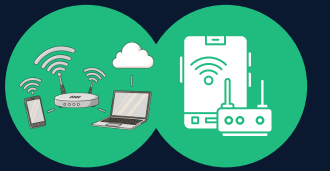
- Misconfigured wireless gateway API endpoint
- API is accessible without proper authentication
- Exposed customer personal information database

Root Cause:

- Inadequate network segmentation
- API is not behind proper access controls
- Wireless gateway exposed to the internet

Lesson: Physical wireless security is insufficient without network-layer access controls

Module 7: Wireless Security Best Practices



7.1 Configuration Hardening

Access Point Setup:

1. Change Default Credentials: Admin username/password immediately
2. Update Firmware: Patch known vulnerabilities
3. Strong Encryption: Minimum WPA2, prefer WPA3 if devices support
4. Strong Passphrase: 20+ character random string (if PSK)
5. Use 802.1X: Enterprise authentication for corporate networks
6. Disable WPS: Wi-Fi Protected Setup is bypassable
7. Hide SSID: Minor security through obscurity
8. Enable PMF: 802.11w management frame protection

7.2 Network Architecture

Segmentation:

- Guest network separate from corporate network (VLAN)
- IoT devices are isolated on a restricted VLAN
- BYOD devices in a separate zone
- Admin network on a secure wired connection only

Access Control:

- Zero Trust principle: verify all devices
- MAC address filtering: not primary control, but supplementary
- IP-based restrictions: limit subnet access
- Firewall rules: restrict east-west traffic

Monitoring:

- Intrusion Detection System (IDS) for wireless
- Continuous monitoring for rogue APs
- Log AP authentication events
- Track signal strength anomalies

7.3 802.11w and Management Frame Protection

Concept: Protects management frames (de-auth, disassoc, action) with encryption and authentication

Enabled Modes:

- Optional: Devices prefer protected but accept unprotected
- Required: Only protected frames accepted
- Disabled: No protection (default on legacy devices)

Implementation:

- Requires AP and all clients support 802.11w
- Enable "Required" mode only when all clients are updated
- Prevents de-authentication attacks
- Breaks some legacy device compatibility

7.4 Regulatory Compliance

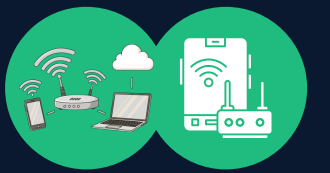
Standards & Frameworks:

- PCI-DSS: Requires WPA2 minimum for payment card processing
- NIST SP 800-153: Government wireless security guidelines
- HIPAA: Healthcare data over wireless must use AES encryption
- IEEE 802.11i: Mandatory in government deployments

Auditing Requirements:

- Annual wireless penetration tests
- Rogue AP detection scans (continuous monitoring preferred)
- Firmware vulnerability assessments
- User training on public Wi-Fi risks

Module 8: Emerging Trends & Future Directions



8.1 Wi-Fi 6E (802.11ax in 6 GHz Band)

Advantages:

- 6 GHz band: 14 additional 80 MHz channels (vs. 3 non-overlapping at 2.4 GHz)
- Massive capacity increase for dense deployments
- Lower latency through less congestion
- Backward compatible with Wi-Fi 6 at 2.4/5 GHz

Security Considerations:

- AFC (Automated Frequency Coordination) is required to avoid spectrum conflicts
- Security of AFC servers becomes critical (potential DDoS target)
- New spectrum = new attack surfaces

8.2 Wi-Fi 7 (802.11be) - 2024 Launch

Key Features:

- Multi-Link Operation (MLO): simultaneous 2.4/5/6 GHz connection
- 46 Gbps theoretical throughput
- Ultra-low latency (< 2ms)
- Improved power efficiency

Security Research Needed:

- MLO handshake security implications
- Increased attack surface with multiple simultaneous links
- Quantum-resistant authentication mechanisms

8.3 Post-Quantum Cryptography

Threat: Quantum computers will break current RSA/ECC encryption

NIST Standards (as of 2023):

- CRYSTALS-Kyber: Lattice-based key encapsulation (selected for standardization)
- CRYSTALS-Dilithium: Lattice-based digital signatures
- Implementation timeline: 2024-2026

Wireless Impact:

- 802.1X certificate authentication will need quantum-safe algorithms
- WPA3 enterprise certificates will require updating
- Legacy devices unable to upgrade pose a security risk

8.4 AI-Driven Security

Emerging Applications:

- Behavioral analysis for rogue AP detection (e.g., Darktrace)
- Anomalous signal pattern recognition
- Predictive threat modeling
- Automated vulnerability patching

Challenges:

- Training data availability
- False positive rates in operational networks
- Privacy implications of behavioral monitoring

Summary



- Evolution of Encryption: RC4 (broken) → TKIP (weak) → AES (strong) → AES-256 with SAE (stronger)
- Authentication: From no auth → shared key → PSK → 802.1X → SAE
- Management Security: From unprotected → 802.11w (optional) → 802.11w (required in WPA3)
- Reconnaissance: Passive discovery → active enumeration → handshake capture → offline cracking
- Defense in Depth: Encryption + authentication + network segmentation + monitoring + incident response

ADDITIONAL RESOURCES



- IEEE 802.11-2020: Wireless LAN Standard
- NIST SP 800-153: Guidelines for Securing Wireless Local Area Networks (WLANs)
- RFC 2289: A One-Time Password System
- RFC 3748: Extensible Authentication Protocol (EAP)
- Vanhoef, M., & Piessens, F. (2017). Key Reinstallation Attacks
- Vanhoef, M. (2019). Dragonblood: Analyzing the Dragonfly Handshake of WPA3
- Wi-Fi Alliance. (2023). Wi-Fi 6E & Wi-Fi 7 Security Specifications

Thank You

ANY QUESTIONS?

